PTB-01 Real Measurement Test Kit

Probe test block, calibration and test procedure — Mold Dimensional Deviation Assessment System, thesis Chapter 4.6

1. Purpose

This kit gives you a part to machine, a calibration method, and a step-by-step test procedure to produce the real-machine results the thesis still needs: probe tip calibration, repeatability, accuracy against a reference instrument, reproducibility after re-clamping, and measuring time. Every test uses the normal app workflow (STEP → G-code + Schema → OpenBuilds Control → Log → Evaluation), so the results also prove the whole system works end to end.

2. What is in the kit

Item	File / location	Use
Test block PTB-01	probe_test_block_PTB01.step	Part to machine and measure (Sheet 1)
Ring gauge model	ring_gauge_D20.step (from make_ring_gauge.py)	Load in the app to calibrate the Effective Tip Ø
Test procedure	this document, sections 4–8	What to run and in what order
Data sheets	this document, section 9	Print and fill in during the tests

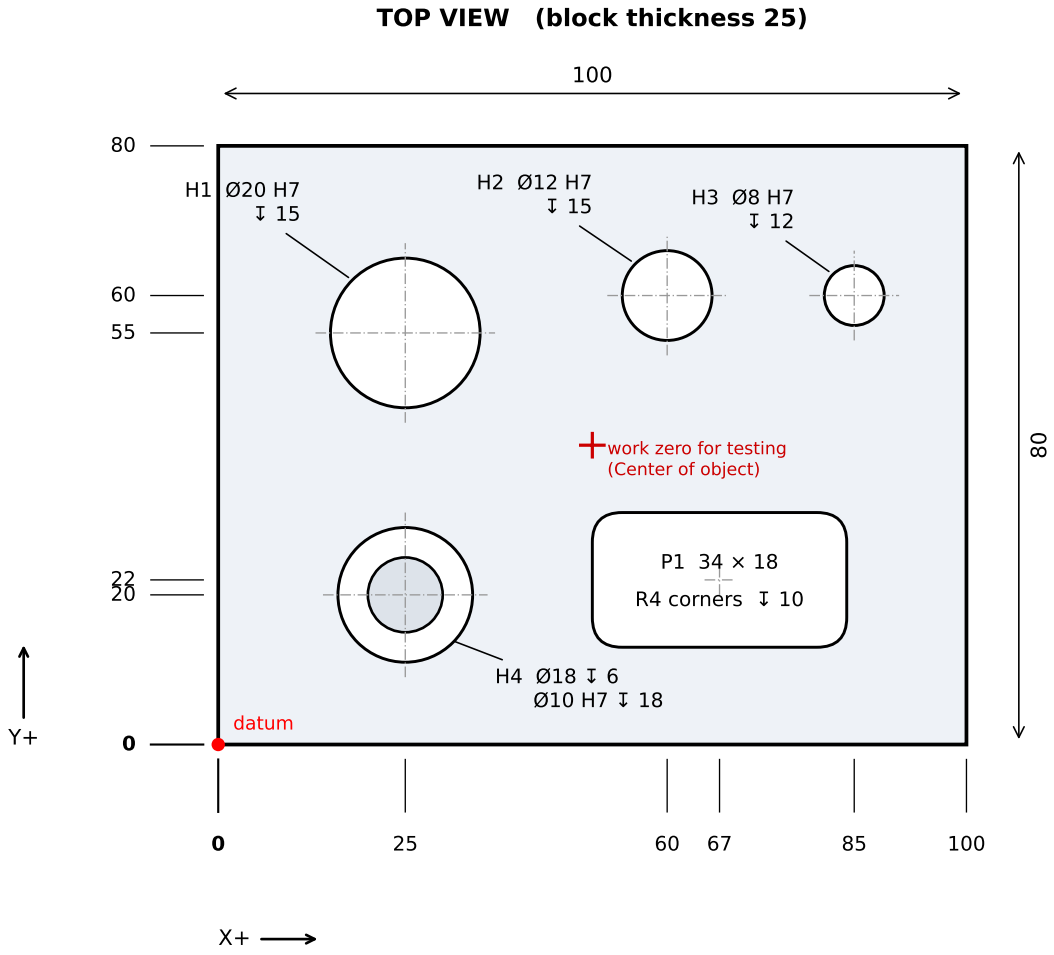
3. Equipment needed

Equipment	Requirement	Why
CNC + V6 touch probe	Stylus 22 mm, ball Ø2 (record actual)	System under test
Ring gauge Ø20	Class X or better, with certificate (any size Ø10–Ø30 works)	Calibrate Effective Tip Ø
Reference instrument for holes	Bore gauge / 3-point internal micrometer (resolution ≤ 0.002), or university CMM	True diameters of H1–H4
Pin gauge set	Ø7.9–Ø8.1 in 0.01 steps	True Ø of H3 (small hole)
Caliper / height gauge	Resolution 0.01	Pocket size, depths, block size
Gauge block / feeler	Known thickness, e.g. 5.000	Set Z0 (probe needs no Z trigger)
Thermometer	±0.5 °C	Record room temperature
Vise or clamps + parallels	—	Hold part and ring gauge rigidly

> Golden rule: never rotate the probe in the spindle and never change the probe feed rate between calibration and the tests. Both change the trigger point (pre-travel), so the calibrated Effective Tip Ø becomes invalid and you must calibrate again.

Sheet 1 — Part Drawing: PTB-01

All dimensions in mm. Datum: lower-left-top corner (X0 Y0 at front-left edge, Z0 = top face).



Feature table

ID	X	Y	Size	Depth	Type
H1	25	55	Ø20 H7	15	round hole, blind
H2	60	60	Ø12 H7	15	round hole, blind
H3	85	60	Ø8 H7	12	round hole, blind
H4	25	20	Ø18 / Ø10 H7	6 / 18	counterbore (2 segments)
P1	67	22	34 × 18, R4	10	rectangular pocket, blind

General notes

- Material: aluminium 6061-T6 (or 7075 / tool steel). Stock squared on all 6 faces, flatness and squareness ≤ 0.02. Outside size tolerance ±0.05.
- H1–H3 and H4 lower bore: drill, then bore or ream to finish (target H7, roundness ≤ 0.01). The exact diameters do NOT need to hit nominal — the true size comes from the reference measurement (section 5).
- Break all edges with a chamfer ≤ 0.3 × 45°. No burrs inside the holes. Wash off coolant/oil.
- Pocket corner R4: finish with an end mill ≤ Ø6. Holes and pocket depths ±0.1.
- CAD file: test_models/probe_test_block_PTB01.step (generated by make_probe_test_block.py).

Making and Reference-Measuring the Block

4. Machining

1. Face and square the stock to 100 × 80 × 25. The outside faces are used to find the work zero, so they must be flat and square.
2. Machine all features from the top in one setup. Use a fresh finishing tool for the last pass in each bore (0.1–0.2 mm radial stock) so walls are smooth and round.
3. Deburr, wash, and let the part reach room temperature (≥ 1 h) before any measurement.
4. You may machine it on your own CNC or outsource it. Either way, the reference measurement below — not the drawing — defines the true size.

5. Reference measurement (do this BEFORE probing)

The app compares against CAD nominal, so its deviation = machining error + measuring error. To judge the measuring system alone you need the true size of each feature, measured with an instrument that is at least ~4× more accurate than the target accuracy (0.01–0.02 mm).

- Measure each round diameter (H1, H2, H3, H4 upper Ø18 and lower Ø10) at the SAME three depths the app will probe (read them from the Customization tab after section 6.3), in two directions: along X (0°) and along Y (90°). Record on Data Sheet B.
- H3 Ø8: use pin gauges — the largest pin that enters freely = diameter (±0.005).
- Pocket P1: length and width with caliper inside jaws or gauge blocks at mid-depth; depth with depth gauge.
- Depths of all features with a depth micrometer or caliper depth rod. Type them into the Evaluation tab (hole card → Caliper depth) — the app compares them with the CAD depth when the probe cannot touch the floor.
- Write down room temperature. Aluminium grows 0.023 µm per mm per °C: a Ø20 hole changes ~0.5 µm per °C, so keep reference and probe measurements within ±2 °C.
- If a CMM is available at the university, measure the whole block there — it gives diameters, centre positions and roundness, which allows the full comparison in section 8.

6. Machine and app setup

6.1 Machine preparation

- Check backlash compensation in OpenBuilds Control and the anti-backlash nuts. Run the axes for 10 minutes to warm up before the first test of the day.
- Mount the probe and do not rotate it again for the whole test series. Check the stylus is tight and the ball is clean.
- Confirm the probe input works: in the console send `?` while touching the ball — the status must show Pn:P.

6.2 Probe profile in the app (Hardware Setting → Probe)

Field	Value
Tip Diameter Ø	actual ball size (e.g. 2.0)
Stylus Length	22
Min. Wall Clearance	0.5
Effective Tip Ø — calibrated	0 (until section 7 is done)

6.3 Probing strategy (use the same for every run)

Feature	Layers	Points / layer	Zigzag
Ring gauge, H1, H2, H3	3	8	off
H4 (each segment)	3	8	off

Feature	Layers	Points / layer	Zigzag
P1	3	12	off

- View: Top, screen rotation 0°. Work zero: Center of object (Export Settings).
- Probe feed: 100 mm/min, overtravel 0.8, backoff 1.2 (defaults). Keep them for ALL runs.
- In the Customization tab check the top layer is ≥ 1.5 mm below the top face (clear of the chamfer) and the bottom layer ≥ 1.5 mm above the floor.
- Evaluation tolerance: 0.05 mm (also re-evaluate at 0.02 and 0.10 for discussion).

Setting the Work Zero with the Probe

Use 'Center of object' as work zero. Finding a centre by touching two opposite faces cancels the ball radius and the pre-travel, so X/Y zero does not depend on the probe calibration. The G-code expects Z0 = top face at the BALL CENTRE.

A. Square the part to the machine X axis

1. Jog the ball in front of the front face (Y−), about 5 mm below the top, near X = 20 of the part. Probe toward +Y and note the Y value of the [PRB:...] line.
2. Repeat near X = 80 (60 mm away). If the two Y values differ by more than 0.02 mm, tap the part and repeat. (0.02 over 60 mm = 0.02° rotation.)

B. X centre — probe the left and right faces

```
; ball ~5 mm left of the left face, ~5 mm below the top, Y near mid-part
G91 G38.2 X10 F100        ; touch left face  -> note PRB X = X1 (machine coords)
G0 X-2                    ; back off
; lift above the part, jog to ~5 mm right of the right face, lower again
G91 G38.2 X-10 F100       ; touch right face -> note PRB X = X2
G0 X2
; lift, then go to the centre in machine coordinates and set X0
G90 G53 G0 X[(X1+X2)/2]   ; type the computed number
G10 L20 P1 X0
```

Bonus cross-check: (X2 − X1) − (block length measured with caliper) = Effective Tip Ø seen from the outside. It should agree with section 7 within ~0.01 mm.

C. Y centre — same with the front and back faces

```
G91 G38.2 Y10 F100   ; front face -> Y1      ...   G91 G38.2 Y-10 F100   ; back face -> Y2
G90 G53 G0 Y[(Y1+Y2)/2]
G10 L20 P1 Y0
```

D. Z zero — top face, ball centre (gauge block — no Z trigger needed)

Many touch probes only trigger from the side, so do NOT probe the top face downward. Use a gauge block (or feeler gauge) of known thickness T instead:

```
; ball above the block centre (solid material at X0 Y0); gauge block T on the top face
; jog Z down in 0.01 mm steps until the gauge block just stops sliding under the ball
G10 L20 P1 Z[T + 1.000]  ; T + ball radius (Ø2 ball), e.g. T = 5.000 -> Z6.000
G91 G0 Z10               ; retract, remove the gauge block
```

Accuracy about ±0.02 mm — enough, because Z0 only sets the layer heights, not the measured diameters. The app also keeps every deepest layer at least 'Min. Floor Clearance' (Hardware Setting, default 1 mm) above the floor or step below, so a small Z0 error cannot push the ball into the floor.

> Always lift Z before moving across the part. The G38.2 distances above are short on purpose — if the probe does not trigger, GRBL raises an alarm instead of crashing. Clear it with $X and check the probe input.

The same procedure is used for the ring gauge, but the X/Y centre is found from the INSIDE: touch −X and +X inside the bore, go to the midpoint, touch −Y and +Y, go to the midpoint, then repeat X once more (the first X chord was not through the centre). Z0 = top face of the ring, set with the gauge block as above.

Test Procedure

7. Calibration — Effective Tip Ø (Data Sheet A)

1. Generate the ring model with the certificate size: python make_ring_gauge.py 20.003 50 15 (bore, outer Ø, height of YOUR gauge).
2. Clamp the ring gauge on parallels, set the work zero at its centre (section 'Setting the Work Zero', last paragraph).
3. In the app load ring_gauge_D20.003.step, Top view, select the hole, 3 layers × 8 points, work zero Center of object, Effective Tip Ø = 0. Export G-code + Schema.
4. Run it in OpenBuilds Control. Save the log. Repeat the run 3 times without touching anything.
5. In the Evaluation tab load Schema + each log. For every layer read 'Ball-centre Ø'. Write the 9 values on Data Sheet A.
6. Effective Tip Ø = Ring Ø − mean(Ball-centre Ø). Enter it in Hardware Setting → Probe → Effective Tip Ø and Apply.
7. Check: re-evaluate the logs — the measured Ø of every layer should now equal the ring Ø within ±0.005. If one direction is always worse, note the lobing (see Roundness).

8. Tests on PTB-01

Test 1 — Repeatability (Data Sheet C)

1. Clamp PTB-01, set work zero (Center of object). Select only H1. Export.
2. Run the same program 10 times in a row without touching the part or the machine. Save 10 logs.
3. Evaluate each log (same Schema). Record measured Ø, centre offset and roundness of each layer.
4. Result: standard deviation s of each quantity. Repeatability = 2s (≈95 %). Target: 2s ≤ 0.02 mm (probe spec 0.01 + machine).

Test 2 — Accuracy vs reference (Data Sheet D)

1. Select all 5 features (H1–H4, P1). Export one program. Run it 5 times. Save 5 logs.
2. For each round feature/segment and layer: mean measured Ø over the 5 runs.
3. Error = mean measured Ø − reference Ø (Data Sheet B, same depth). Target after calibration: |error| ≤ 0.03 mm.
4. Also record the app's per-point max offset vs CAD and the pass/fail verdict at tolerance 0.05 — then check whether the verdict agrees with the reference (if the reference itself is > 0.05 from nominal, the app SHOULD fail that feature).
5. P1 pocket: record max offset per layer; compare the measured walls with the caliper width (qualitative — the app has no pocket size fit yet).

Test 3 — Reproducibility / re-setup (Data Sheet E)

1. Unclamp the part, re-clamp it in a slightly different place, square it, set the work zero again from scratch. Run the full program once. Do this 3 times.
2. Compare the diameters with Test 2 (diameter should not change much) and the centre positions (these show the work-zero setting error).

Test 4 — Measuring time (Data Sheet F)

- Time each stage: app setup → export, work-zero setting, machine run, evaluation. Compare with measuring the same features by hand (bore gauge + caliper).

Optional Test 5 — Deliberate error

- Shift the work zero by +0.10 mm in X (G10 L20 P1 X-0.1 at the zero point) and run H1 once. The centre offset must show ≈0.10 mm and the diameter must stay the same — proves the evaluation separates position error from size error.

Analysis and reporting

```
mean        x_mean = Σ xi / n
std. dev.   s  = sqrt( Σ (xi − x_mean)² / (n − 1) )
repeatability  = 2·s                       (Test 1, ~95 % of repeated readings)
bias / error   = x_mean − x_ref                 (Test 2)
expanded uncertainty (simple)  U ≈ 2·sqrt( s1² + s3² + u_ref² )
      s1 = Test 1 std. dev., s3 = Test 3 std. dev., u_ref = reference instrument std. unc.
```

- Thesis Ch. 4.6 tables = Data Sheets A, C, D, E, F (mean ± s). Plot measured − reference per feature as a bar chart with ±U error bars.
- Ch. 4.7 discussion: compare bias of the large (Ø20) vs small (Ø8) hole — a size-dependent error points to tip calibration; a constant centre shift points to work zero; high roundness on a gauge points to probe lobing or backlash.
- Agree the target values (0.02 / 0.03 mm) with the advisor before testing.

9. Data Sheets

Date: _____ Operator: _____ Room temp: _____ °C Probe feed: _____ mm/min

A. Tip calibration — ring gauge (Ring Ø cert: _____ Ball Ø nominal: _____)

Run	Layer 1 Ball-centre Ø	Layer 2 Ball-centre Ø	Layer 3 Ball-centre Ø	Run mean
1				
2				
3				
Mean				

Effective Tip Ø = Ring Ø − mean = _____

Check after entering: measured Ø L1 _____ L2 _____ L3 _____

B. Reference measurements (instrument: _____)

Feature	Nominal	Depth layer	Ø at 0°	Ø at 90°	Mean Ø (ref)
H1	Ø20	L1			
H1	Ø20	L2			
H1	Ø20	L3			
H2	Ø12	L1			
H2	Ø12	L2			
H2	Ø12	L3			
H3	Ø8	L1			
H3	Ø8	L2			
H3	Ø8	L3			
H4 top	Ø18	L1			
H4 top	Ø18	L2			
H4 top	Ø18	L3			
H4 bottom	Ø10	L1			
H4 bottom	Ø10	L2			
H4 bottom	Ø10	L3			
P1 length	34	mid		—	
P1 width	18	mid		—	

C. Test 1 — Repeatability, H1 (10 runs, part not touched)

Run	Ø L1	Ø L2	Ø L3	Centre off. L2	Roundness L2	Max offset
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
Mean						
s						
2s						

D. Test 2 — Accuracy (5 runs, all features) tolerance used: _____

Feature / layer	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	s	Ref Ø	Error
H1 L1									
H1 L2									
H1 L3									
H2 L1									
H2 L2									
H2 L3									
H3 L1									
H3 L2									
H3 L3									
H4 top L1									
H4 top L2									
H4 top L3									
H4 bot L1									
H4 bot L2									
H4 bot L3									
P1 max off.									
Verdict (P/F)									

E. Test 3 — Reproducibility (re-clamp + new work zero each time)

Feature (L2)	Setup 1 Ø	Setup 2 Ø	Setup 3 Ø	s (Ø)	Centre X,Y setup 1	Centre X,Y setup 2	Centre X,Y setup 3
H1							
H2							
H3							
H4 top							
H4 bot							

F. Test 4 — Time

Stage	Run 1 (min)	Run 2 (min)	Run 3 (min)	Mean
App: load STEP → export				
Set work zero				
Machine run (all features)				
Evaluation				
TOTAL (system)				
Manual (bore gauge + caliper)				

G. Notes / problems